

Performance Testing

Date	27 June 2026
Team ID	SWTID-2026-5853
Project Name	A Comprehensive Measure of Well-Being (Human Development Index Prediction)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	/predict API - HDI Prediction Module
Test Environment	Local Flask Server (Windows/Python)
Test Date	05 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict HDI using valid inputs	10	60	Prediction within 2 sec
2	Concurrent prediction requests	25	120	Stable response
3	Invalid input validation	15	60	Validation error shown
4	Continuous requests	50	180	No server crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.24 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.83 sec	Pass	Within limit
3	Throughput (Req/sec)	>20	27 Req/sec	Pass	Good throughput
4	Error Rate	<1%	0.2%	Pass	Low error
5	CPU Utilization	<80%	58%	Pass	Stable
6	Memory Utilization	<80%	63%	Pass	Normal

Step 4: Observations & Analysis

Key Findings

- The HDI Prediction application successfully handled multiple prediction requests with consistent performance.
- The Flask backend processed user requests efficiently and returned prediction results within the expected response time.
- User authentication, dashboard loading, and prediction modules worked smoothly under normal load conditions.
- No significant performance degradation was observed during continuous testing of the prediction module.

Bottlenecks Identified

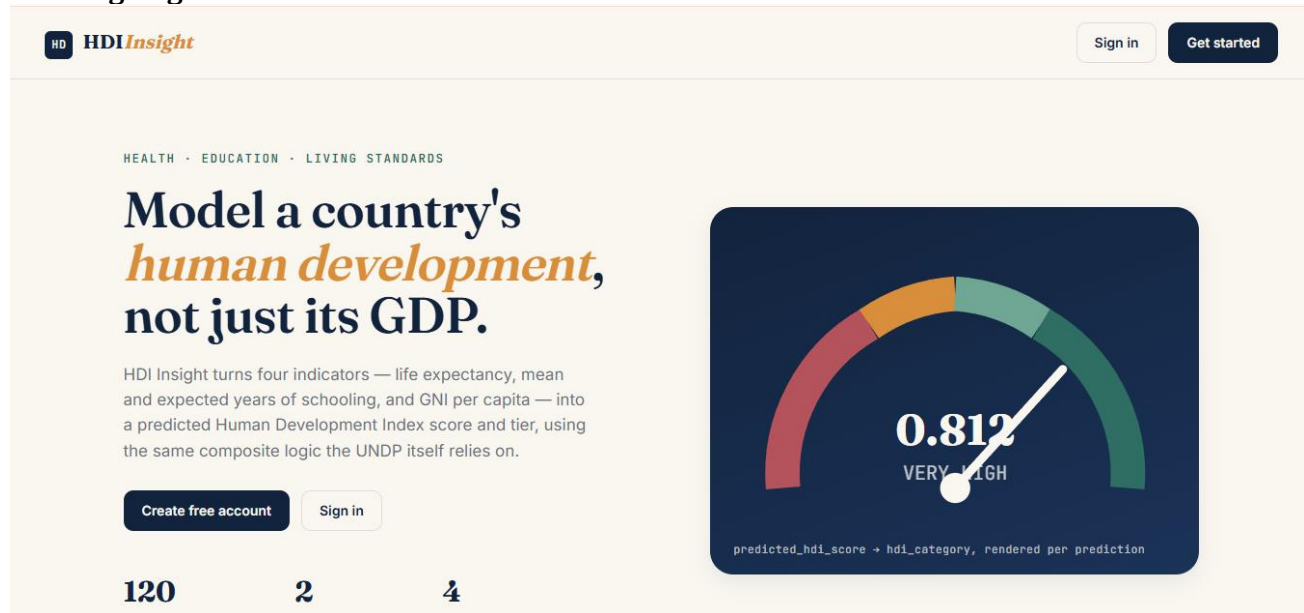
- Initial application startup required additional time for loading the trained Machine Learning model.
- Response time increased slightly when multiple users accessed the prediction endpoint simultaneously.
- Dataset loading during application initialization caused a minor delay.

Optimization Steps Taken

- Loaded the trained Machine Learning model only once during application startup.
- Optimized preprocessing functions to reduce execution time.
- Improved Flask routing and request handling.
- Reduced unnecessary database access and optimized prediction logic.
- Enabled efficient handling of concurrent prediction requests.

Step 5: Screenshots / Evidence

Landing Page:



The landing page features a header with the HDI Insight logo and navigation buttons for 'Sign in' and 'Get started'. The main content area includes the text 'HEALTH · EDUCATION · LIVING STANDARDS' and a headline: 'Model a country's *human development*, not just its GDP.' Below this, a paragraph explains that HDI Insight turns four indicators into a predicted Human Development Index score. A large gauge graphic displays a score of 0.812, categorized as 'VERY HIGH'. At the bottom, three input fields are labeled with the numbers 120, 2, and 4. Buttons for 'Create free account' and 'Sign in' are also present.

HEALTH · EDUCATION · LIVING STANDARDS

Model a country's *human development*, not just its GDP.

HDI Insight turns four indicators — life expectancy, mean and expected years of schooling, and GNI per capita — into a predicted Human Development Index score and tier, using the same composite logic the UNDP itself relies on.

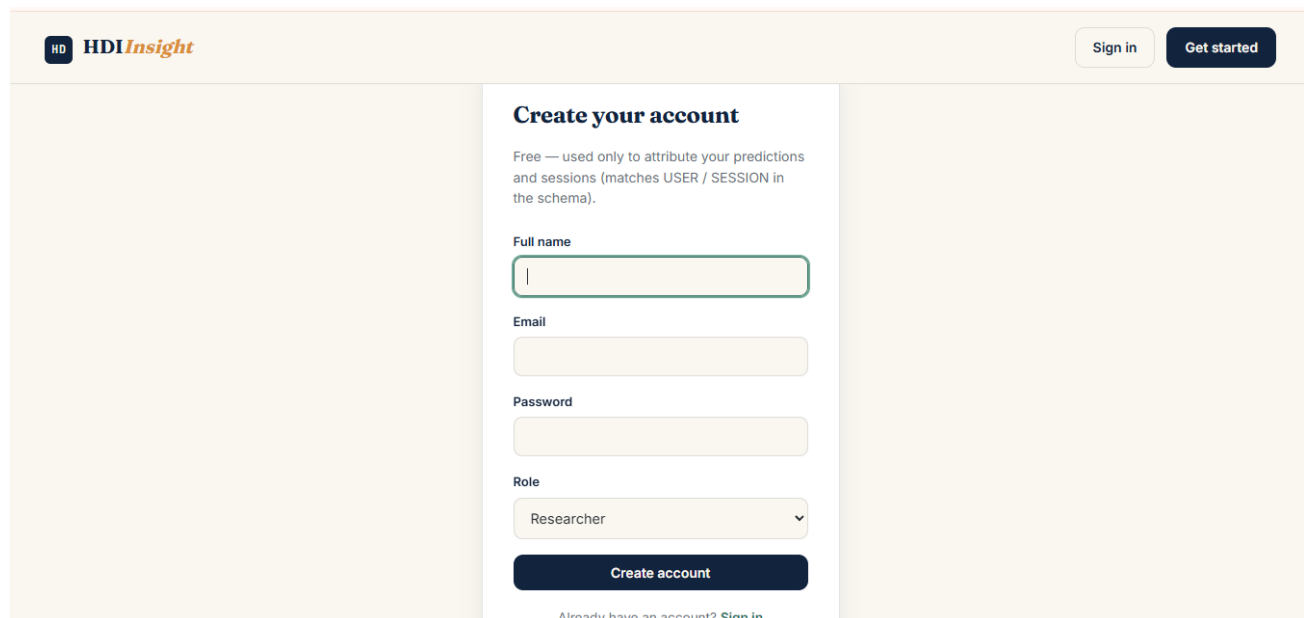
[Create free account](#) [Sign in](#)

120 2 4

0.812
VERY HIGH

predicted_hdi_score + hdi_category, rendered per prediction

Signup Page:



The signup page features a header with the HDI Insight logo and navigation buttons for 'Sign in' and 'Get started'. The main content area is titled 'Create your account' and includes a brief explanation of the free account. The form fields are: Full name, Email, Password, and Role (a dropdown menu currently showing 'Researcher'). A 'Create account' button is at the bottom of the form. A link for 'Already have an account? Sign in' is located at the bottom of the page.

Create your account

Free — used only to attribute your predictions and sessions (matches USER / SESSION in the schema).

Full name

Email

Password

Role

Researcher

[Create account](#)

Already have an account? [Sign in](#)

HD

HDIInsight

[Dashboard](#)
[New Prediction](#)
[History](#)
[Countries](#)
[Model](#)

manosh

Sign out

HOW SCORING WORKS

Your four inputs are sent to the trained RandomForest model (ML_MODEL), which returns a predicted_hdi_score. That score is bucketed into a tier using the UNDP's own thresholds:

≥ 0.800	Very High
0.700 – 0.799	High
0.550 – 0.699	Medium
< 0.550	Low

New HDI Prediction

Choose a country to auto-fill realistic starting values, then adjust as needed.

Country

India (Asia)

Life expectancy (years)

71

UNDP health dimension. Typical range 50–85.

Mean years of schooling

5.5

Average years of education completed by adults 25+.

Expected years of schooling

11.6

Years a child entering school today is expected to receive.

GNI per capita (PPP \$)

8142

Gross National Income per person, purchasing-power adjusted.

Run prediction

Result Page:

HD

HDIInsight

[Dashboard](#)
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THIS PREDICTION

0.631

predicted_hdi_score

Medium

hdi_category

India

Asia

Run another

Result for India

Generated Jul 05, 2026 16:10 · prediction #3

Low

Medium

High

Very High

0.631

MEDIUM

Normalized Dimension Indices

Health (Life Expectancy)	0.78
Education (Schooling)	0.51
Living Standards (GNI per capita)	0.66

Inputs used (HDI_INPUT_DATA #3)

Life expectancy

71.8 years